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STANDARDS

IEEE Standard for
Local and Metropolitan Area Networks—

Port-Based Network Access Control—Corrigendum 1

IEEE Computer Society

Developed by the
LAN/MAN Standards Committee

IEEE Std 802.1X™-2020/Cor 1-2026
(Corrigendum to IEEE Std 802.1X™-2020)

IEEE

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IEEE Standard for Local and Metropolitan Area Networks—

Port-Based Network Access Control—Corrigendum 1

Developed by the

LAN/MAN Standards Committee
of the
IEEE Computer Society

Approved 25 September 2026

IEEE SA Standards Board

Abstract: This corrigendum to IEEE Std 802.1X-2020 corrects the when statement for the augment of the interfaces container in the ieee802-dot1x YANG module to avoid unduly restricting the applicability of the ieee802-dot1x YANG module to specific interface types.

Keywords: access control, authentication, authorization, controlled port, corrigendum, EAP, EAPOL, IEEE 802.1X, key agreement, LANs, local area networks, MACsec, MACsec Key Agreement, MAC security, MAC Service, MANs, metropolitan area networks, MKA, port-based network access control, secure association, security, service access point, uncontrolled port

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Introduction

This introduction is not part of IEEE Std 802.1X™-2020/Cor 1-2026, IEEE Standard for Local and Metropolitan Area Networks—Port-Based Network Access Control—Corrigendum 1.

This corrigendum to IEEE Std 802.1X corrects the when statement for the augment of the interfaces container in the ieee802-dot1x YANG module to avoid limiting the latter's applicability. The YANG modules in this document and their previous revisions, available at https://1.ieee802.org/yang-modules/, can be compared to identify their differences.

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IEEE Standard for
Local and Metropolitan Area Networks—

Port-Based Network Access Control—Corrigendum 1

[This corrigendum is based on IEEE Std 802.1X™-2020]

NOTE—The editing instructions contained in this corrigendum define how to merge the material contained therein into the existing base standard to form the comprehensive standard.

The editing instructions are shown in ***bold italics***. Four editing instructions are used: change, delete, insert, and replace. ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change and describes what is being changed by using ~~strikethrough~~ (to remove old material) and underscore (to add new material). ***Delete*** removes existing material. ***Insert*** adds new material without disturbing the existing material. Deletions and insertions may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is used to make changes in figures or equations by removing the existing figure or equation and replacing it with a new one. Editing instructions, change markings, and this note will not be carried over into future editions because the changes will be incorporated into the base standard.

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14. YANG Data Model

14.5 Definition of the IEEE 802.1X YANG data model [6, 7, 8]

14.5.2 ieee802-dot1x-types YANG module definition

Delete the YANG module in 14.5.2.

Insert the following YANG module in 14.5.2:

```
module ieee802-dot1x-types {
  yang-version "1.1";
  namespace "urn:ieee:std:802.1X:yang:ieee802-dot1x-types";
  prefix "dot1x-types";

  organization
    "Institute of Electrical and Electronics Engineers";

  contact
    "WG-URL: http://www.ieee802.org/1
    WG-EMail: stds-802-1-L@ieee.org

    Contact: IEEE 802.1 Working Group Chair
    Postal: C/O IEEE 802.1 Working Group
            IEEE Standards Association
            445 Hoes Lane
            Piscataway
            NJ 08854
            USA

    E-mail: stds-802-1-chairs@ieee.org";

  description
    "Port-based network access control allows a network administrator
    to restrict the use of IEEE 802 LAN service access points (ports)
    to secure communication between authenticated and authorized
    devices. IEEE Std 802.1X specifies an architecture, functional
    elements, and protocols that support mutual authentication
    between the clients of ports attached to the same LAN and secure
    communication between the ports. The following control allows a
    port to be reinitialized, terminating (and potentially
    restarting) authentication exchanges and MKA operation, based on
    a data model described in a set of YANG modules.";

  revision 2026-06-16 {
    description
      "Added statement to indicate YANG version 1.1";
    reference
      "IEEE Std 802.1X-2020, Port-Based Network Access Control.";
  }

  revision 2020-02-18 {
    description
      "Updated Contact information.";
    reference
      "IEEE Std 802.1X-2020, Port-Based Network Access Control.";
  }

  revision 2019-05-28 {
    description
      "Updates based upon comment resolution on draft
```

[6] Copyright release for YANG: Users of this standard may freely reproduce the YANG modules contained in this clause so that they can be used for their intended purpose.

[7] An ASCII version of each YANG module is attached to the PDF of this standard and can also be obtained from the IEEE 802.1 Website at https://1.ieee802.org/yang-modules/.

[8] References in this standard's YANG module definitions are not clickable, as each module has been incorporated unchanged after development and verification using YANG tools.

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```
    D1.0 of P802.1X-Rev.";
  reference
    "IEEE Std 802.1X-2020, Port-Based Network Access Control.";
}

/* ---------------------------------------------
 * Type definitions used by dot1X YANG module
 * ---------------------------------------------
 */

typedef pae-nid {
  type string {
    length "0..100";
  }
  description
    "Network Identity, which is a UTF-8 string identifying a
    network or network service.";
  reference
    "IEEE 802.1X-2020 Clause 3, Clause 10.1, Clause 12.6";
}

typedef pae-session-user-name {
  type string {
    length "0..253";
  }
  description
    "Session user name, which is a UTF-8 string, representing the
    identity of the peer Supplicant.";
  reference
    "IEEE 802.1X-2020 Clause 12.5.1";
}

typedef pae-session-id {
  type string {
    length "3..253";
  }
  description
    "Session Identifier, which is a UTF-8 string, uniquely
    identifying the session within the context of the PAE's
    system.";
  reference
    "IEEE 802.1X-2020 Clause 12.5.1";
}

typedef pae-nid-capabilities {
  type bits {
    bit eap {
      position 0;
      description
        "EAP";
    }
    bit eapMka {
      position 1;
      description
        "EAP + MKA";
    }
    bit eapMkaMacSec {
      position 2;
      description
        "EAP + MKA + MACsec";
    }
    bit mka {
      position 3;
      description
        "MKA";
    }
    bit mkaMacSec {
      position 4;
      description
        "MKA + MACsec";
    }
    bit higherLayer {
```


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```
      position 5;
      description
        "Higher Layer (WebAuth)";
    }
    bit higherLayerFallback {
      position 6;
      description
        "Higher Layer Fallback (WebAuth)";
    }
    bit vendorSpecific {
      position 7;
      description
        "Vendor specific authentication mechanisms";
    }
  }
  description
    "Authentication and protection capabilities supported for the
    NID. Indicates the combinations of authentication and
    protection capabilities supported for the NID. Any set of these
    combinations can be supported.";
  reference
    "IEEE 802.1X-2020 Clause 10.1, Clause 11.12.3";
}

typedef pae-access-status {
  type enumeration {
    enum no-access {
      description
        "Other than to authentication services, and to services
        announced as available in the absence of authentication
        (unauthenticated).";
    }
    enum remedial-access {
      description
        "The access granted is severely limited, possibly to
        remedial services.";
    }
    enum restricted-access {
      description
        "The Controlled Port is operational, but restrictions have
        been applied by the network that can limit access to some
        resources.";
    }
    enum expected-access {
      description
        "The Controlled Port is operational, and access provided is
        as expected for successful authentication and authorization
        for the NID.";
    }
  }
  description
    "Indicates the transmitter's Controlled Port operational status
    and current level of access resulting from authentication and
    the consequent authorization controls applied by that port's
    clients.";
  reference
    "IEEE 802.1X-2020 Clause 10.4, Clause 12.5";
}

typedef mka-kn {
  type uint32;
  description
    "Indicates a Key Number (KN) used in MKA. It is assigned by
    the Key Server (sequentially beginning with 1).";
  reference
    "IEEE 802.1X-2020 Clause 9.8, Clause 9.16";
}

typedef mka-an {
  type uint32;
  description
    "A number that is concatenated with a MACsec Secure Channel
```


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```
      Identifier to identify a Secure Association. Indicates an
      Association Number (AN) assigned by the Key Server for use with
      the key number for transmission.";
    reference
      "IEEE 802.1X-2020 Clause 9.8, Clause 9.16";
  }

  typedef pae-ckn {
    type string {
      length "1..32";
    }
    description
      "Indicates the CAK name to identify the Connectivity
      Association Key (CAK) which is the root key in the MACsec Key
      Agreement key hierarchy. All potential members of the CA use
      the same CKN.";
    reference
      "IEEE 802.1X-2020 Clause 9.3.1, Clause 6.2";
  }

  typedef pae-kmd {
    type string {
      length "0..253";
    }
    description
      "A Key Management Domain (KMD). A string of up to 253 UTF-8
      characters that names the transmitting authenticator's key
      management domain.";
    reference
      "IEEE Clause 12.6";
  }

  typedef pae-auth-data {
    type string;
    description
      "Authorization data associated with the CAK.";
    reference
      "IEEE 802.1X-2020 Clause 9.16";
  }

  typedef sci-list-entry {
    type string {
      length "8";
    }
    description
      "8 octet string, where the first 6 octets represents the MAC
      Address (in canonical format), and the next 2 octets represents
      the Port Identifier.";
    reference
      "IEEE 802.1AE Clause 7.1.2, Clause 10.7.1";
  }

  typedef pae-if-index {
    type int32 {
      range "1..2147483647";
    }
    description
        "The interface index value represented by this interface.";
  }

} // ieee802-dot1x-types
```

14.5.3 ieee802-dot1x YANG module definition

Delete the YANG module in 14.5.3.

Insert the following YANG module in 14.5.3:

```
module ieee802-dot1x {
  yang-version "1.1";
```


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```
namespace "urn:ieee:std:802.1X:yang:ieee802-dot1x";
prefix "dot1x";

import ieee802-types { prefix "ieee"; }
import ietf-yang-types { prefix "yang"; }
import ietf-interfaces { prefix "if"; }
import ietf-system { prefix "sys"; }
import iana-if-type { prefix "ianaift"; }
import ieee802-dot1x-types { prefix "dot1x-types"; }

organization
  "Institute of Electrical and Electronics Engineers";

contact
  "WG-URL: http://www.ieee802.org/1
  WG-EMail: stds-802-1-L@ieee.org

  Contact: IEEE 802.1 Working Group Chair
  Postal: C/O IEEE 802.1 Working Group
          IEEE Standards Association
          445 Hoes Lane
          Piscataway
          NJ 08854
          USA

  E-mail: stds-802-1-chairs@ieee.org";

description
  "Port-based network access control allows a network administrator
  to restrict the use of IEEE 802 LAN service access points (ports)
  to secure communication between authenticated and authorized
  devices. IEEE Std 802.1X specifies an architecture, functional
  elements, and protocols that support mutual authentication
  between the clients of ports attached to the same LAN and secure
  communication between the ports. The following control allows a
  port to be reinitialized, terminating (and potentially
  restarting) authentication exchanges and MKA operation, based on
  a data model described in a set of YANG modules.";

revision 2026-06-16 {
  description
    "Added statement to indicate YANG version 1.1 and added
        derived-from-or-self function to the when statement
       for the augment of the interfaces container.";
  reference
    "IEEE Std 802.1X-2020, Port-Based Network Access Control.";
}

revision 2020-02-18 {
  description
    "Updated Contact information.";
  reference
    "IEEE Std 802.1X-2020, Port-Based Network Access Control.";
}

revision 2019-06-12 {
  description
    "Updates based on comment resolution of the WG ballot of
    P802.1X-Rev/D1.0.";
  reference
    "IEEE Std 802.1X-2020, Port-Based Network Access Control.";
}

grouping nid-group {
  description
    "The PAE NID Group configuration and operational information.";
  list pae-nid-group {
    key "nid";
    description
      "A list that contains the configuration and operational
      nodes for the network announcement information for the
      Logon Process.";
```


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```
  leaf nid {
    type dot1x-types:pae-nid;
    description
      "Identification of the network or network service.";
    reference
      "IEEE 802.1X-2020 Clause 12.5";
  }
  leaf use-eap {
    type enumeration {
      enum never {
        description
          "Never.";
      }
      enum immediate {
        description
          "Immediately, concurrently with the use of MKA with any
          cached CAK(s).";
      }
      enum mka-fail {
        description
          "Not until MKA has failed, if a prior CAK has been
          cached.";
      }
    }
    default "immediate";
    description
      "Determines when the Logon Process will initiate EAP, if
      the Supplicant and or Authenticator are enabled, and takes
      one of the above values.";
    reference
      "IEEE 802.1X-2020 Clause 12.5";
  }
  leaf unauth-allowed {
    type enumeration {
      enum never {
        description
          "Never.";
      }
      enum immediate {
        description
          "Immediately, independently of any current or future
          attempts to authenticate using the PAE or MKA.";
      }
      enum auth-fail {
        description
          "Not until an attempt has been made to authenticate
          using EAP, unless neither the supplicant nor the
          authenticator is enabled, and MKA has attempted to use
          any cached CAK (unless the KaY is not enabled).";
      }
    }
    default "immediate";
    description
      "Determines when the Logon Process will tell the CP state
      machine to provide unauthenticated connectivity, and takes
      one of the above values.";
    reference
      "IEEE 802.1X-2020 Clause 12.5";
  }
  leaf unsecure-allowed {
    type enumeration {
      enum never {
        description
          "Never.";
      }
      enum immediate {
        description
          "Immediately, to provide connectivity concurrently with
          the use of MKA with any CAK acquired through EAP.";
      }
      enum mka-fail {
        description
```


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```
            "Not until MKA has failed, or is not enabled.";
          }
          enum mka-server {
            description
              "Only if directed by the MKA server.";
          }
        }
        default "immediate";
        description
          "Determines when the Logon Process will tell the CP state
          machine to provide authenticated but unsecured
          connectivity, takes one of the above values.";
        reference
          "IEEE 802.1X-2020 Clause 12.5";
      }
      leaf unauthenticated-access {
        type enumeration {
          enum no-access {
            description
              "Other than to authentication services.";
          }
          enum fallback-access {
            description
              "Limited access can be provided after authentication
              failure.";
          }
          enum limited-access {
            description
              "Immediate limited access is available without
              authentication.";
          }
          enum open-access {
            description
              "Immediate access is available without
              authentication.";
          }
        }
        default "no-access";
        description
          "Unauthenticated access capabilities provided by the NID.";
        reference
          "IEEE 802.1X-2020 Clause 10.1";
      }
      leaf access-capabilities {
        type dot1x-types:pae-nid-capabilities;
        description
          "Authentication and protection capabilities supported for
          the NID.";
        reference
          "IEEE 802.1X-2020 Clause 10.1";
      }

      leaf kmd {
        type dot1x-types:pae-kmd;
        config false;
        description
          "The Key Management Domain for the NID.";
        reference
          "IEEE 802.1X-2020 Clause 10.4";
      }
    }
  }

  grouping port-capabilities {
    description
      "Per port PAE feature capabilities.";
    leaf supp {
      type boolean;
      description
        "Indicates if PACP EAP Supplicant is supported.";
      reference
        "IEEE 802.1X-2020 Clause 12.9.2";
```


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```
  }
  leaf auth {
    type boolean;
    description
      "Indicates if PACP EAP Authenticator is supported.";
    reference
      "IEEE 802.1X-2020 Clause 12.9.2";
  }
  leaf mka {
    type boolean;
    description
      "Indicates if MKA is supported.";
    reference
      "IEEE 802.1X-2020 Clause 12.9.2";
  }
  leaf macsec {
    type boolean;
    description
      "Indicates if MACsec on the Controlled port is supported.";
    reference
      "IEEE 802.1X-2020 Clause 12.9.2";
  }
  leaf announcements {
    type boolean;
    description
      "Indicates if the ability to send EAPOL announcements is
      supported.";
    reference
      "IEEE 802.1X-2020 Clause 12.9.2";
  }
  leaf listener {
    type boolean;
    description
      "Indicates if the ability to use received EAPOL
      announcements is supported.";
    reference
      "IEEE 802.1X-2020 Clause 12.9.2";
  }
  leaf virtual-ports {
    type boolean;
    description
      "Indicates if virtual ports for a real port is supported.";
    reference
      "IEEE 802.1X-2020 Clause 12.9.2";
  }
  leaf in-service-upgrades {
    type boolean;
    description
      "Indicates if MKA in-service upgrades is supported.";
    reference
      "IEEE 802.1X-2020 Clause 12.9.2";
  }
}

/* ------------------------------------------------
 * Configuration objects used by 802.1X YANG module
 * ------------------------------------------------
 */
augment "/sys:system" {
  description
    "Augment system with 802.1X PAE System specific configuration
    nodes.";
  container pae-system {
    description
      "Contains all 802.1X PAE System specific related
      configuration and operational data.";
    leaf name {
      type string {
        length "1..255";
      }
      description
        "The name which uniquely identifies the PAE System.";
```


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```
  }
  leaf system-access-control {
    type enumeration {
      enum disabled {
        description
          "Deletes any virtual ports previously instantiated, and
          terminates authentication exchanges and MKA
          operation.";
      }
      enum enabled {
        description
          "Enables PAE system access control.";
      }
    }
    description
      "Setting this control to disabled deletes any virtual ports
      previously instantiated, and terminates authentication
      exchanges and MKA operation. Each real port PAE behaves as
      if enabledVirtualPorts was clear, the PAEs Supplicant,
      Authenticator, and KaY as if their enabled controls were
      clear, and Logon Process(es) as if unauthAllowed was
      Immediate. Announcements can be transmitted (subject to
      other controls), both periodically and in response to
      announcement requests (conveyed by EAPOL-Starts or
      EAPOL-Announcement-Reqs) but are sent with a single NID
      Set, with a null NID, and the Access Information TLV (and
      no other) with an pae-access-status of No Access,
      accessRequested false, OpenAccess, and no
      accessCapabilities. The control variable settings for each
      real port PAE are unaffected, and will be used once
      systemAccessControl is set to enabled.";
    reference
      "IEEE 802.1X-2020 Clause 12.9.1";
  }
  leaf system-announcements {
    type enumeration {
      enum disabled {
        description
          "Causes each PAE to behave as if enabled were clear
          for the PAE's Announcement functionality.";
      }
      enum enabled {
        description
          "Enables PAE system announcements.";
      }
    }
    description
      "Setting this control to Disabled causes each PAE to behave
      as if enabled were clear for the PAE's Announcement
      functionality. The independent controls for each PAE apply
      if systemAnnouncements is Enabled.";
    reference
      "IEEE 802.1X-2020 Clause 12.9.1";
  }
  leaf eapol-protocol-version {
    type uint8;
    config false;
    description
      "The EAPOL protocol version for this system.";
    reference
      "IEEE 802.1X-2020 Clause 12.9.1, Clause 11.3";
  }
  leaf mka-version {
    type uint8;
    config false;
    description
      "The MKA protocol version for this system.";
    reference
      "IEEE 802.1X-2020 Clause 12.9.1, Clause 11.3";
  }
  leaf-list pae {
    type if:interface-ref;
```


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```
      config false;
      description
        "List of PAE references.";
    }
  }
}

/*
 *  Port Authentication Entity (PAE) Nodes
 */
augment "/if:interfaces/if:interface" {
  when "derived-from-or-self(if:type,'ianaift:ethernetCsmacd') or " +
       "derived-from-or-self(if:type,'ianaift:ilan') or " +
       "derived-from-or-self(if:type,'ianaift:macSecControlledIF') or " +
       "derived-from-or-self(if:type,'ianaift:ptm') or " +
       "derived-from-or-self(if:type,'ianaift:bridge')" {
    description
      "Applies to Ethernet, bridge, MACsec controlled, LAN, and
       point-to-multipoint interfaces supporting port-based
       access control.";
  }
  description
    "Augment interface model with PAE configuration and
     operational nodes.";
  reference
    "IEEE 802.1AE Clause 11.7 and IEEE 802.1X-2020 Clause 6.5 and
    Clause 13.3.2";
  container pae {
    description
      "Contains PAE configuration and operational related nodes.";
    leaf pae-system {
      type leafref {
        path "/sys:system/dot1x:pae-system/dot1x:name";
      }
      description
        "The PAE system that this PAE is a member of.";
    }
    leaf vp-enable {
      when "../port-type = 'real-port' and
            ../port-capabilities/virtual-ports = 'true'" {
        description
          "Applies when port is Real Port and virtual port
          capabilities are supported.";
      }
      type boolean;
      default "false";
      description
        "A real port's PAE may be configured to create virtual
        ports to support multi-access LANs provided that MKA and
        MACsec operation is enabled for that port.";
      reference
        "IEEE 802.1X-2020 Clause 12.7";
    }
    container port-capabilities {
      description
        "Per port PAE feature capabilities.";
      uses port-capabilities;
    }

    leaf port-name {
      type if:interface-ref;
      config false;
      description
        "Each PAE is uniquely identified by a port name.";
      }
    leaf port-number {
      type dot1x-types:pae-if-index;
      config false;
      description
        "Each PAE is uniquely identified by a port number. The
        port number used is unique amongst all port names for the
        system, and directly or indirectly identifies the
```


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```
      Uncontrolled Port that supports the PAE. If the PAE has
      been dynamically instantiated to support an existing or
      potential virtual port, this portNumber, the
      uncontrolledPortNumber and the controlledPortNumber are
      allocated by the real ports PAE, and this portNumber is the
      uncontrolledPortNumber. If the PAE supports a real port,
      this portNumber is the commonPortNumber for the associated
      PAC or SecY.";
    reference
      "IEEE 802.1X-2020 Clause 12.9.2";
    }
  leaf controlled-port-name {
    type if:interface-ref;
    config false;
    description
      "Each PAE is uniquely identified by a port name.";
  }
  leaf controlled-port-number {
    type dot1x-types:pae-if-index;
    config false;
    description
      "The port for the associated PAC or SecYs Controlled
      Port.";
    reference
      "IEEE 802.1X-2020 Clause 12.9.2";
  }
  leaf uncontrolled-port-name {
    type if:interface-ref;
    config false;
    description
      "The uncontrolled port name reference.";
  }
  leaf uncontrolled-port-number {
    type dot1x-types:pae-if-index;
    config false;
    description
      "The port for the associated PAC or SecYs Uncontrolled
      Port.";
    reference
      "IEEE 802.1X-2020 Clause 12.9.2";
  }
  leaf common-port-name {
    type if:interface-ref;
    config false;
    description
      "The common port name reference.";
  }
  leaf common-port-number {
    type dot1x-types:pae-if-index;
    config false;
    description
      "The port for the associated PAC or SecYs Common Port. All
      the virtual ports created for a given real port share the
      same Common Port and commonPortNumber.";
    reference
      "IEEE 802.1X-2020 Clause 12.9.2";
  }
  leaf port-type {
    type enumeration {
      enum real-port {
        description
          "Real Port type.";
      }
      enum virtual-port {
        description
          "Virtual Port type.";
      }
    }
    //config false;
    description
      "The port type of the PAE.";
    reference
```


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```
      "IEEE 802.1X-2020 Clause 12.9.2";
    }
    container virtual-port {
      when "../port-capabilities/virtual-ports = 'true'" {
        description
          "Applies when the virtual ports port capability is
          supported.";
      }
      config false;
      description
        "Contains Virtual Port operational state information.";
      leaf max {
        when "../../port-type = 'real-port'" {
          description
            "Applies when Port is a Real Port.";
        }
        type uint32;
        description
          "The guaranteed maximum number of virtual ports.";
        reference
          "IEEE 802.1X-2020 Clause 12.9.2";
      }
      leaf current {
        when "../../port-type = 'real-port'" {
          description
            "Applies when Port is a Real Port.";
        }
        type yang:gauge32;
        description
          "The current number of virtual ports.";
        reference
          "IEEE 802.1X-2020 Clause 12.9.2";
      }
      leaf start {
        when "../../port-type = 'virtual-port'" {
          description
            "Applies when Port is a Virtual Port.";
        }
        type boolean;
        description
          "Set if the virtual port was created by receipt of an
          EAPOL-Start frame.";
        reference
          "IEEE 802.1X-2020 Clause 12.9.7";
      }
      leaf peer-address {
        when "../../port-type = 'virtual-port'" {
          description
            "Applies when Port is a Virtual Port.";
        }
        type ieee:mac-address;
        description
          "The source MAC Address of the EAPOL-Start (if vpStart is
          set).";
        reference
          "IEEE 802.1X-2020 Clause 12.9.7";
      }
    }

    container supplicant {
      when "../port-type = 'real-port' and
            ../port-capabilities/supp = 'true'" {
        description
          "Applies to Real Port when supplicant port capabilities
          are supported.";
      }
      description
        "Contains the configuration nodes for the Supplicant PAE
        associated with each port.";
      leaf held-period {
        type uint16;
        units seconds;
```


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```
      default "60";
      description
        "The initial value of the timer used to impose a wait
        period after a failed authentication attempt, before
        another attempt is permitted.";
      reference
        "IEEE 802.1X-2020 Clause 8.6";
    }
    leaf retry-max {
      type uint32;
      default "2";
      description
        "Specifies the maximum number of re-authentication
        attempts on an authenticator port before port is
        unauthorized.";
      reference
        "IEEE 802.1X-2020 Clause 8.7";
    }

    leaf enabled {
      type boolean;
      config false;
      description
        "Set by PACP if the PAE can provide authentication. Will
        be FALSE if the Port is not enabled, if the functionality
        provided by the PAE is not available, or not implemented,
        or the control variable enable has been cleared by
        management, e.g. because the application scenario
        authenticates a user and there is no user logged on.";
      reference
        "IEEE 802.1X-2020 Clause 8.4";
    }
    leaf authenticate {
      type boolean;
      config false;
      description
        "Set by the PAE client to request authentication, and
        allows reauthentication while set. Cleared by the client
        to revoke authentication. To enable authentication the
        client also needs to clear failed (if set).";
      reference
        "IEEE 802.1X-2020 Clause 8.4";
    }
    leaf authenticated {
      type boolean;
      config false;
      description
        "Set by PACP if the PAE is currently authenticated, and
        cleared if the authentication fails or is revoked.";
      reference
        "IEEE 802.1X-2020 Clause 8.4";
    }
    leaf failed {
      type boolean;
      config false;
      description
        "Set by PACP if the authentication has failed or has been
        terminated. The cause could be a Fail returned by EAP,
        either immediately or following a reauthentication, an
        excessive number of attempts to authenticate (either
        immediately or upon reauthentication), or the client
        deasserting authenticate. The PACP will clear
        authenticated as well as setting failed. Any ongoing
        authentication exchange will be terminated (by the state
        machines) if enable becomes FALSE and enabled will be
        cleared, but failed will not be set.";
      reference
        "IEEE 802.1X-2020 Clause 8.4";
    }
  }

  container authenticator {
```


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```
when "../port-capabilities/auth = 'true'" {
  description
    "Applies when the Authenticator is supported.";
}
description
  "Contains configuration nodes for the Authenticator PAE
  associated with each port.";
leaf quiet-period {
  type uint16;
  units seconds;
  default "60";
  description
    "Number of seconds that the authenticator remains in the quiet
    state following a failed authentication exchange with the
    supplicant.";
  reference
    "IEEE 802.1X-2020 Clause 8.6, Figure 12-3";
}
leaf reauth-period {
  type uint32;
  units seconds;
  default "3600";
  description
    "This object indicates the time period of the
    reauthentication to the supplicant.";
  reference
    "IEEE 802.1X-2020 Clause 8.6, Figure 12-3";
}
leaf reauth-enable {
  type boolean;
  default "false";
  description
    "Re-authentication is enabled or not.";
  reference
    "IEEE 802.1X-2020 Clause 5.8 and 8.9";
}
leaf retry-max {
  type uint32;
  default "2";
  description
    "Specifies the maximum number of re-authentication
    attempts on an authenticator port before port is
    unauthorized.";
  reference
    "IEEE 802.1X-2020 Clause 8.9";
}

leaf enabled {
  type boolean;
  config false;
  description
    "Set by PACP if the PAE can provide authentication. Will
    be FALSE if the Port is not enabled, if the functionality
    provided by the PAE is not available, or not implemented,
    or the control variable enable has been cleared by
    management, e.g. because the application scenario
    authenticates a user and there is no user logged on.";
  reference
    "IEEE 802.1X-2020 Clause 8.4";
}
leaf authenticate {
  type boolean;
  config false;
  description
    "Set by the PAE client to request authentication, and
    allows reauthentication while set. Cleared by the client
    to revoke authentication. To enable authentication the
    client also needs to clear failed (if set).";
  reference
    "IEEE 802.1X-2020 Clause 8.4";
}
leaf authenticated {
```


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```
      type boolean;
      config false;
      description
        "Set by PACP if the PAE is currently authenticated, and
        cleared if the authentication fails or is revoked.";
      reference
        "IEEE 802.1X-2020 Clause 8.4";
    }
    leaf failed {
      type boolean;
      config false;
      description
        "Set by PACP if the authentication has failed or has been
        terminated. The cause could be a Fail returned by EAP,
        either immediately or following a reauthentication, an
        excessive number of attempts to authenticate (either
        immediately or upon reauthentication), or the client
        deasserting authenticate. The PACP will clear
        authenticated as well as setting failed. Any ongoing
        authentication exchange will be terminated (by the state
        machines) if enable becomes FALSE and enabled will be
        cleared, but failed will not be set.";
      reference
        "IEEE 802.1X-2020 Clause 8.4";
    }
  }

  container kay {
    when "../port-capabilities/mka = 'true'" {
      description
        "Applies when the MKA port capability is supported.";
    }
    description
      "Contains configuration system level information for each
      Interface supported by the KaY (Key Aggreement Entity).";
    leaf enable {
      type boolean;
      default "false";
      description
        "Set by management to enable (clear to disable) the use
        of MKA.";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }
    container actor {
      description
        "Contains configuration and operational nodes
        associated with the actor";
      leaf priority {
        type uint8;
        description
          "The Key Server Priority for all the ports actors.";
        reference
          "IEEE 802.1X-2020 Clause 9.16";
      }
      leaf sci {
        type dot1x-types:sci-list-entry;
        config false;
        description
          "The SCI assigned by the system to the port (applies
          to all the ports actors).";
        reference
          "IEEE 802.1X-2020 Clause 9.16";
      }
    }
    container key-server {
      description
        "Contains configuration and operational nodes
        associated with the key
        server.";
      leaf priority {
        type uint8;
```


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```
      description
        "The Key Server Priority for the Key Server for the
        principal actor. Matches the actorPriority if the
        actor is the Key Server";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }
    leaf sci {
      type dot1x-types:sci-list-entry;
      config false;
      description
        "The SCI for Key Server for the principal actor. Null
        if there is no principal actor, or that actor has no
        live peers. Matches the actorSCI if the actor is the
        Key Server.";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }
  }
  container group {
    description
      "Contains configuration nodes associated with the
      group.";
    leaf join {
      type boolean;
      default "true";
      description
        "Set if the KaY will accept Group CAKs distributed by
        MKA.";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }
    leaf form {
      type boolean;
      default "false";
      description
        "Set if the KaY will attempt to use point-to-point CAs
        to distribute a Group CAK, if its principal actor is
        the Key Server for all the point-to-point CAs.";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }
    leaf new {
      type boolean;
      default "false";
      description
        "Set by management if a new Group CAK is to be
        distributed, if the principal actor is the Key Server
        for all point-to-point CAs. Cleared by the KaY when
        distribution is complete.";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }
  }

  container macsec {
    when "../../port-capabilities/macsec = 'true'" {
      description
        "Applies when the MACsec port capability is
        supported.";
    }
    description
      "Contains configuration and operational nodes
      associated with macsec.";
    leaf capable {
      type boolean;
      description
        "Set if MACsec is implemented.";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }
    leaf desired {
```


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```
      type boolean;
      default "true";
      description
        "Set if the participant desires MACsec frame protection.";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }

    leaf protect {
      type boolean;
      config false;
      description
        "As used by the CP state machine, see 12.4.";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }
    leaf validate {
      type boolean;
      config false;
      description
        "As used by the CP state machine, see 12.4.";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }
    leaf replay-protect {
      type boolean;
      config false;
      description
        "As used by the CP state machine, see 12.4.";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }
  }
  leaf suspend-on-request {
    type boolean;
    default "true";
    description
      "Set by management to allow the KaYs principal actor to
      initiate a suspension if it is the Key Server and another
      participant has requested a suspension.";
  }
  leaf suspend-for {
    type uint8;
    default "0";
    description
      "Set by management to a non-zero number of seconds
      between 1 and MKA Suspension Limit to initiate a
      suspension (9.18) of that duration (if the KaYs principal
      actor is the Key Server) or to request a suspension
      (otherwise).";
    reference
      "IEEE 802.1X-2020 Clause 9.18";
  }

  leaf suspended-while {
    type uint8;
    config false;
    description
      "Read by management to determine if a suspension is in
      progress and (when available) to discover the remaining
      duration of that suspension";
    reference
      "IEEE 802.1X-2020 Clause 9.18";
  }
  leaf active {
    type boolean;
    config false;
    description
      "Set if there is at least one active actor, transmitting
      MKPDUs.";
    reference
      "IEEE 802.1X-2020 Clause 9.16";
```


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```
    }
    leaf authenticated {
      type boolean;
      config false;
      description
        "Set if the principal actor, i.e. the participant that
        has the highest priority Key Server and one or more live
        peers, has determined that Controlled Port communication
        should proceed without MACsec.";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }
    leaf secured {
      type boolean;
      config false;
      description
        "Set if the principal actor has determined that
        communication should use MACsec.";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }
    leaf failed {
      type boolean;
      config false;
      description
        "Cleared when authenticated or secured are set, set if
        the latter are clear and MKA Life Time has elapsed since
        an MKA participant was last created.";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }
    container key-number {
        config false;
      description
        "Contains operation state nodes for Key Numbers.";
      leaf tx {
        type dot1x-types:mka-kn;
        description
          "The Key Number assigned by the Key Server to the SAK
          currently being used for transmission. Null if MACsec
          is not being used.";
        reference
          "IEEE 802.1X-2020 Clause 9.16";
      }
      leaf rx {
        type dot1x-types:mka-kn;
        description
          "The Key Number assigned by the Key Server to the
          oldest SAK currently being used for reception. The same
          as txKN if a single SAK is currently in use (as will
          most often be the case). Null if MACsec is not being
          used.";
        reference
          "IEEE 802.1X-2020 Clause 9.16";
      }
    }
    container association-number {
        config false;
      description
        "Contains operation state nodes for Association
        Numbers.";
      leaf tx {
        type dot1x-types:mka-an;
        description
          "The Association Number assigned by the Key Server for
          use with txKN. Zero if MACsec is not in use.";
        reference
          "IEEE 802.1X-2020 Clause 9.16";
      }
      leaf rx {
        type dot1x-types:mka-an;
        description
```


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```
        "The Association Number assigned by the Key Server for
        use with rxKN. The same as txAN if a single SAK is
        currently in use. Zero if MACsec is not in use.";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }
  }

  list participants {
    key "participant";
    description
      "Contains list of configuration and operational nodes
      for each MKA participant supported by the KaY MKA
      entity.";
    leaf participant {
      type uint32;
      description
        "Key into Participants list.";
    }
    leaf cached {
      type boolean;
      description
        "Set by the KaY if the participants parameters are
        cached. If set, cached can be cleared by management to
        remove the participant from the cache.";
    }
    leaf active {
      type boolean;
      default "false";
      description
        "Set if the participant is active, i.e., is currently
        transmitting periodic MKPDUs.";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }
    leaf retain {
      type boolean;
      default "false";
      description
        "Set by management to retain the participant in the
        cache, even if the KaY would normally remove it (due to
        lack of use for example).";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }
    leaf activate {
      type enumeration {
        enum default {
          description
            "The participant is from cached entries created by
            the KaY as part of normal operation, without
            explicit management, and is activated according to
            the implementation dependent policies of the KaY.";
        }
        enum disabled {
          description
            "The participant allows the cache information to be
            retained, but disabled for indefinite period.";
            }
        enum on-oper-up {
          description
            "Causing the participant to be activated when the
            PAEs part is activated, and therefore when the SecY
            or PACs Common Port becomes operational.";
        }
        enum always {
          description
            "Causing the participant to remain active all the
            time, even in the continued absence of partners.";
        }
      }
      default "default";
```


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```
      description
        "Controls when the participant is activated. Cached
        entries created by the KaY as part of normal operation,
        without explicit management, have the value Default,
        and are activated according to the implementation
        dependent policies of the KaY. This variable can be
        set to any of its values by management. Disabled allows
        the cache entry to be retained, but disabled for an
        indefinite period. OnOperUp causes the participant to
        be activated when the PAEs port (and therefore when the
        SecY or PACs Common Port becomes MAC_Operational).
        Always causes the participant to remain active all the
        time, even in the continued absence of partners. If the
        value is changed to Disabled or OnOperUp, the
        participant ceases operation immediately and receipt of
        MKPDUs with a matching CKN during a subsequent period
        of twice MKA Life Time will not cause the participant
        to become active once more.";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }

    container peers {
      config false;
      description
        "Contains operational state nodes associated with the
        Peers.";
      leaf-list live {
        type dot1x-types:sci-list-entry;
          description
            "A list of the SCIs of the participants live
            peers.";
          reference
            "IEEE 802.1X-2020 Clause 9.16";
      }
      leaf-list potential {
        type dot1x-types:sci-list-entry;
        description
          "A list of the SCIs of the participants potential
          peers.";
        reference
          "IEEE 802.1X-2020 Clause 9.16";
      }
    }
    leaf ckn {
      type dot1x-types:pae-ckn;
      config false;
      description
        "The secure Connectivity Association Key Name for the
        participant.";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }
    leaf kmd {
      type dot1x-types:pae-kmd;
      config false;
      description
        "The Key Management Domain for the participant.";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }
    leaf nid {
      type dot1x-types:pae-nid;
      config false;
      description
        "The NID for the participant.";
      reference
        "IEEE 802.1X-2020 Clause 9.16";
    }
    leaf auth-data {
      type dot1x-types:pae-auth-data;
      config false;
```


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```
        description
          "Authorization data associated with the secure
          Connectivity Association Key.";
        reference
          "IEEE 802.1X-2020 Clause 9.16";
      }
      leaf principal {
        type boolean;
        config false;
        description
          "Set if the participant is currently the principal
          actor.";
        reference
          "IEEE 802.1X-2020 Clause 9.16";
      }
      leaf dist-ckn {
        type dot1x-types:pae-ckn;
        config false;
        description
          "The CKN for the last CAK distributed (either by the
          actor or one of its partners). Null if this participant
          has not been used to distribute a CAK.";
        reference
          "IEEE 802.1X-2020 Clause 9.16";
      }
    }
  }

  container logon-nid {
    description
      "Contains the configuration and operational related NID
        information for the Logon Process. The Logon Process may
        use Network Identifiers (NIDs) to manage its use of
        authentication credentials, cached CAKs, and
        announcements.";
    leaf selected {
      type dot1x-types:pae-nid;
      description
        "The NID currently configured for use by an access
        controlled port when transmitting EAPOL-Start frames.
        Defaults to the null NID.";
      reference
        "IEEE 802.1X-2020 Clause 12.5";
    }
    uses nid-group;

    leaf connected {
      type dot1x-types:pae-nid;
      config false;
      description
        "The NID associated with the current connectivity
        (possibly unauthenticated) provided by the operation of
        the CP state machine.";
      reference
        "IEEE 802.1X-2020 Clause 12.5";
    }
    leaf requested {
      type dot1x-types:pae-nid;
      config false;
      description
        "The NID marked as Access requested in announcements, as
        determined from EAPOL-Start frames. Defaults to the
        selectedNID.";
      reference
        "IEEE 802.1X-2020 Clause 12.5";
    }
  }

  container announcer {
    when "../port-capabilities/announcements = 'true'" {
      description
        "Applies when the Announcements port capabilities are
```


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```
        supported.";
    }
    description
      "Contains the configuration related Announcer
      information.";
    leaf enable {
      type boolean;
      default "false";
      description
        "A boolean indicating if the announcer is enabled or
        not.";
      reference
        "IEEE 802.1X-2020 Clause 10.4";
    }
    list announce {
      key "announces";
      description
        "Contains the configuration related status information
        that the Announcers announce in the network announcement
        of the PAE system.";
      leaf announces {
        type uint32;
        description
          "Key into Announce list.";
      }
      uses nid-group;

      leaf nid {
        type dot1x-types:pae-nid;
        config false;
        description
          "The NID information to identify a received network
          announcement for the PAE.";
        reference
          "IEEE 802.1X-2020 Clause 10.4";
      }
      leaf access-status {
        type dot1x-types:pae-access-status;
        config false;
        description
          "Access Status reflects connectivity as a result of
          authentication attempts, and might be set directly by
          the system or configured by AAA protocols.";
        reference
          "IEEE 802.1X-2020 Clause 10.4, Clause 12.5";
      }
    }
  }

  container listener {
    when "../port-capabilities/listener = 'true'" {
      description
        "Applies when the Listener port capability is
        supported.";
    }
    description
      "Contains the configuration and operational Listener
        node related information.";
    leaf enable {
      type boolean;
      default "false";
      description
        "A boolean indicating if the listener is enabled or
        not.";
      reference
        "IEEE 802.1X-2020 Clause 10.4";
    }

    list announcement {
      key "announcements";
      config false;
      description
```


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```
      "A list containing the operational status information
      that the Listeners receive in the network announcement of
      the PAE system.";
    leaf announcements {
      type uint32;
      description
        "The key into the list of Announce nodes.";
    }
    leaf nid {
      type dot1x-types:pae-nid;
      description
        "The NID information to identify a received network
        announcement for the PAE.";
      reference
        "IEEE 802.1X-2020 Clause 10.4";
    }
    leaf kmd {
      type dot1x-types:pae-kmd;
      description
        "The KMD information for this received network
        announcement of the PAE.";
      reference
        "IEEE 802.1X-2020 Clause 10.4";
    }
    leaf specific {
      type boolean;
      description
        "This object indicates the received announcement
        information was specific to the receiving PAE, not
        generic for all systems attached to the LAN.";
      reference
        "IEEE 802.1X-2020 Clause 10.4";
    }
    leaf access-status {
      type dot1x-types:pae-access-status;
      description
        "The object information reflects connectivity as a
        result of authentication attempts for this received
        network announcement of the PAE.";
      reference
        "IEEE 802.1X-2020 Clause 10.4";
    }
    leaf requested-nid {
      type boolean;
      description
        "The authenticated access has been requested for this
        particular NID or not.";
      reference
        "IEEE 802.1X-2020 Clause 10.4";
    }
    leaf unauthenticated-access {
      type dot1x-types:pae-access-status;
      description
        "The access capability of the ports clients without
        authentication in this received network announcement of
        the PAE";
      reference
        "IEEE 802.1X-2020 Clause 10.4";
    }
    leaf access-capabilities {
      type dot1x-types:pae-nid-capabilities;
      description
        "The authentication and protection capabilities
        supported for the NID.";
      reference
        "IEEE 802.1X-2020 Clause 10.4";
    }
    list cipher-suites {
      key "index";
      description
        "A table contains the Cipher Suites information that
        the Listeners receive in the network announcement of
```


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```
        the PAE system.";
      reference
        "IEEE 802.1X-2020 Clause 10.4";
      leaf index {
        type uint16;
        description
          "Key into cipher suite entry.";
      }
      leaf cipherSuite {
        type string;
        description
          "cipher Suite identifier.";
      }
      leaf cipherSuiteCapability {
        type uint32;
        description
          "Cipher Suite capability.";
      }
    }
  }
}

container eapol-statistics {
  config false;
  description
    "Contains operational EAPOL statistics.";
  leaf invalid-eapol-frame-rx {
    when "../../port-type = 'real-port'" {
      description
        "Applies when port is Real Port.";
    }
    type yang:counter32;
    description
      "The number of invalid EAPOL frames of any type that
      have been received by this PAE.";
    reference
      "IEEE 802.1X-2020 Clause 12.8.1";
  }
  leaf eap-length-error-frames-rx {
    when "../../port-type = 'real-port'" {
      description
        "Applies when port is Real Port.";
    }
    type yang:counter32;
    description
      "The number of EAPOL frames that the Packet Body Length
      does not match a Packet Body that is contained within the
      octets of the received EAPOL MPDU in this PAE.";
    reference
      "IEEE 802.1X-2020 Clause 12.8.1";
  }
  leaf eapol-announcements-rx {
    when "../../port-type = 'real-port'" {
      description
        "Applies when port is Real Port.";
    }
    type yang:counter32;
    description
      "The number of EAPOL-Announcement frames that have been
      received by this PAE";
    reference
      "IEEE 802.1X-2020 Clause 12.8.1";
  }
  leaf eapol-announce-reqs-rx {
    when "../../port-type = 'real-port'" {
      description
        "Applies when port is Real Port.";
    }
    type yang:counter32;
    description
      "The number of EAPOL-Announcement-Req frames that have
      been received by this PAE.";
```


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```
      reference
        "IEEE 802.1X-2020 Clause 12.8.1";
    }
    leaf eapol-port-unavailable {
      when "../../port-type = 'real-port' and
            ../../port-capabilities/virtual-ports = 'true'" {
        description
          "Applies when port is Real Port and when the virtual
          ports capability is supported.";
      }
      type yang:counter32;
      description
        "The number of EAPOL frames that are discarded because
        their processing would require the creation of a virtual
        port, for which there are inadequate or constrained
        resources, or an existing virtual port and no such port
        currently exists. If virtual port is not supported, this
        object should be always 0.";
      reference
        "IEEE 802.1X-2020 Clause 12.8.1";
    }
    leaf eapol-start-frames-rx {
      type yang:counter32;
      description
        "The number of EAPOL-Start frames that have been received
        by this PAE";
      reference
        "IEEE 802.1X-2020 Clause 12.8.1";
    }
    leaf eapol-eap-frames-rx {
      type yang:counter32;
      description
        "The number of EAPOL-EAP frames that have been received
        by this PAE.";
      reference
        "IEEE 802.1X-2020 Clause 12.8.1";
    }
    leaf eapol-logoff-frames-rx {
      type yang:counter32;
      description
        "The number of EAPOL-Logoff frames that have been
        received by this PAE.";
      reference
        "IEEE 802.1X-2020 Clause 12.8.1";
    }
    leaf eapol-mk-no-cfn {
      type yang:counter32;
      description
        "The number of MKPDUs received with MKA not enabled or
        CKN not recognized in this PAE.";
      reference
        "IEEE 802.1X-2020 Clause 12.8.1";
    }
    leaf eapol-mk-invalid-frames-rx {
      type yang:counter32;
      description
        "The number of MKPDUs failing in message authentication
        on receipt process in this PAE.";
      reference
        "IEEE 802.1X-2020 Clause 12.8.1";
    }
    leaf last-eapol-frame-source {
      when "../../port-type = 'real-port'" {
        description
          "Applies when port is Real Port.";
      }
      type ieee:mac-address;
      description
        "The source MAC address of last received EAPOL frame by
        this PAE.";
      reference
        "IEEE 802.1X-2020 Clause 12.8.2";
```


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```
  }
  leaf last-eapol-frame-version {
    type uint8;
    description
      "The version of last received EAPOL frame by this PAE.";
    reference
      "IEEE 802.1X-2020 Clause 12.8.2";
  }
  leaf eapol-supp-eap-frames-tx {
    when "../../port-type = 'real-port'" {
      description
        "Applies when port is Real Port.";
    }
    type yang:counter32;
    description
      "The number of EAPOL-EAP frames that have been
      transmitted by the supplicant of this PAE.";
    reference
      "IEEE 802.1X-2020 Clause 12.8.3";
  }
  leaf eapol-logoff-frames-tx {
    when "../../port-type = 'real-port'" {
      description
        "Applies when port is Real Port.";
    }
    type yang:counter32;
    description
      "The number of EAPOL-Logoff frames that have been
      transmitted by this PAE.";
    reference
      "IEEE 802.1X-2020 Clause 12.8.3";
  }
  leaf eapol-announcements-tx {
    when "../../port-type = 'real-port'" {
      description
        "Applies when port is Real Port.";
    }
    type yang:counter32;
    description
      "The number of EAPOL-Announcement frames that have been
      transmitted by this PAE.";
    reference
      "IEEE 802.1X-2020 Clause 12.8.3";
  }
  leaf eapol-announce-reqs-tx {
    when "../../port-type = 'real-port'" {
      description
        "Applies when port is Real Port.";
    }
    type yang:counter32;
    description
      "The number of EAPOL-Announcement-Req frames that have
      been transmitted by this PAE.";
    reference
      "IEEE 802.1X-2020 Clause 12.8.3";
  }
  leaf eapol-start-frames-tx {
    type yang:counter32;
    description
      "The number of EAPOL-Start frames that have been
      transmitted by this PAE.";
    reference
      "IEEE 802.1X-2020 Clause 12.8.3";
  }
  leaf eapol-auth-eap-frames-tx {
    type yang:counter32;
    description
      "The number of EAPOL-EAP frames that have been
      transmitted by the authenticator of this PAE.";
    reference
      "IEEE 802.1X-2020 Clause 12.8.3";
  }
```


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```
    leaf eapol-mka-frames-tx {
      type yang:counter32;
      description
        "The number of EAPOL-MKA frames with no CKN information
        that have been transmitted by this PAE.";
      reference
        "IEEE 802.1X-2020 Clause 12.8.3";
    }
  }

  container logon-process {
    description
      "Contains configuration and operational system level
        information for each port to support the Logon Process(es)
        status information.";
    leaf logon {
      type boolean;
      default "false";
      description
        "A boolean indicating if the logon-process is enabled or
        not.";
      reference
        "IEEE 802.1X-2020 Clause 12.5";
    }

    leaf connect {
      type enumeration {
        enum pending {
          description
            "Prevent connectivity by clearing the
            controlledPortEnabled parameter.";
        }
        enum unauthenticated {
          description
            "Provide unsecured connectivity, setting
            controlledPortEnabled.";
        }
        enum authenticated {
          description
            "Provide unsecured connectivity with authorization
            data, setting controlledPortEnabled.";
        }
        enum secure {
          description
            "Provide secure connectivity, using SAKs provided by
            the KaY (when available) and setting
            controlledPortEnabled when those keys are installed
            and in use, as specified in detail by the CP state
            machine.";
        }
      }
      config false;
      description
        "The Logon Process sets this variable to one of the
        above values.";
      reference
        "IEEE 802.1X-2020 Clause 12.3";
    }
    leaf port-valid {
      type boolean;
      config false;
      description
        "Set if Controlled Port communication is secured as
        specified by the MACsec control macsecProtect.";
      reference
        "IEEE 802.1X-2020 Clause 12.3";
    }
    list session-statistics {
      key "session-id";
      config false;
      description
        "Contains operational state nodes associated with the
```


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```
        session statistics.";
      leaf session-id {
        type dot1x-types:pae-session-id;
        description
          "Key into list of session statistics.";
        reference
          "IEEE 802.1X-2020 Clause 12.5.1";
      }
      leaf user-name {
        type dot1x-types:pae-session-user-name;
        description
          "User name of the session.";
        reference
          "IEEE 802.1X-2020 Clause 12.5.1";
      }
      leaf octets-rx {
        type yang:counter64;
        description
          "The number of octets received in this session of this
          PAE.";
        reference
          "IEEE 802.1X-2020 Clause 12.5.1";
      }
      leaf octets-tx {
        type yang:counter64;
        description
          "The number of octets transmitted in this session of
          this PAE.";
        reference
          "IEEE 802.1X-2020 Clause 12.5.1";
      }
      leaf frames-rx {
        type yang:counter64;
        description
          "The number of packets received in this session of
          this PAE.";
        reference
          "IEEE 802.1X-2020 Clause 12.5.1";
      }
      leaf frames-tx {
        type yang:counter64;
        description
          "The number of packets transmitted in this session of
          this PAE.";
        reference
          "IEEE 802.1X-2020 Clause 12.5.1";
      }
      leaf time {
        type uint32;
        units "seconds";
        description
          "Session Time. The duration of the session in
          seconds.";
        reference
          "IEEE 802.1X-2020 Clause 12.5.1";
      }
      leaf terminate-cause {
        type enumeration {
          enum common_port_MAC_operational_false {
            description
              "Common Port for this PAE is not operational.";
          }
          enum system_access_control_disabled {
            description
              "The system-access-control node of the pae-system
              is disabled or initialization process of this PAE
              is invoked.";
          }
          enum eapol_logoff_rx {
            description
              "The PAE has received EAPOL-Logoff frame.";
          }
```


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```
            enum eap_reauthentication_failure {
              description
                "EAP reauthentication has failed.";
              }
              enum mka-failure_termination {
                description
                  "MKA failure or other MKA termination.";
              }
              enum new_session-beginning {
                description
                  "New session beginning.";
              }
              enum not_terminated_yet {
                description
                  "Not Terminated Yet.";
              }
            }
            description
              "The reason for the session termination.";
            reference
              "IEEE 802.1X-2020 Clause 12.5.1";
          }
        }
      }
    }
  }

  container nid-group {
    description
      "Contains both configuration and operational state nodes
      associated with the PAE NID group.";
    uses nid-group;
  }

}
```


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